Exam 1
CpE 3150-Digital System Design

Name:_____

Student #:_____

This exam is an individual activity. The exam is open textbook and other class materials posted on Canvas for this course. Other outside materials are NOT permitted. Answers given that are identified as coming from outside sources or as shared work with another classmate will treated as academic dishonesty. All problems with answers identified as dishonest will be scored as 0 points, and there may be a referral to the ECE Department Chair for academic dishonesty.

Show all work. Partial credit will be given based on work shown.

I completed this exam on my own following the guidelines above and did not obtain or provide assistance to anyone else taking this exam. This includes not seeking online assistance from others.

_____ _____

Signature Date

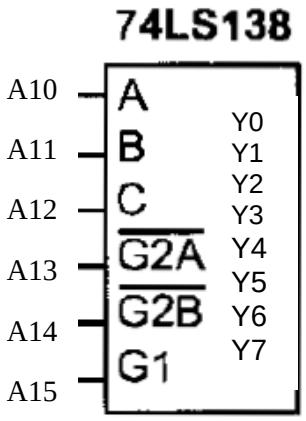
(30 points) Part 1 of the Exam is a quiz (Exam 1: Part 1) on Canvas. This part of the exam consists of True/False, Multiple Choice, and Matching problems. No work is required to be shown for this part of the exam. You may submit any work for this part of the exam on separate sheet(s) with the problem(s) labeled.

(**70 points**) Part 2 of the Exam is given as follows. Showing your work is necessary for partial credit. You may print out this part of the exam and write your solutions, scan them, and submit the scanned document on Canvas for Exam 1. You may also provide your solutions (with work shown) in this Word document and submit on Canvas.

Both parts of the exam are subject to the integrity statement that you are signing above. Both parts of the exam must be completed (part 1 on Canvas) and submitted on Canvas (part 2 here) by the exam deadline. Best wishes with the exam!

Exam 1: Part 2

1. Given the microprocessor configuration below. Answer the following questions.



a. **(5 Points)** Find the address range for Y3 in binary (table) and hex below.

Y3

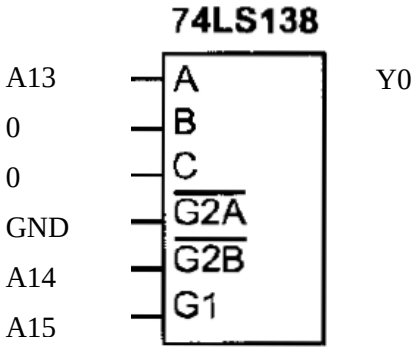
	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
Start	**1**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
End	**1**	**0**	**0**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

Hex address range:

0x8C00-0x8FFF

b. **(6 Points)** Draw the connections for the address lines (A15-A0) on the 74138 below to generate an address range of 0xA000-0xBFFF. Hint: determine the address lines that need to be used for the decoder inputs and enable pins to generate the address range assuming that the remaining address lines are connected to the device being accessed. As a reference to help in this process, draw the address table based on the address range specified. Specify which output of the decoder is activated.

Many possible answers but A15 =1, A14 = 0, A13=1, other decoder inputs need to be specified to satisfy G1, G2A', G2B' constraints and CBA values chosen (meeting constraints) select the decoder output



	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
Start	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
End	**1**	**0**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**

2. For the ATmega32 program below, answer the following questions.

Number of Machine Cycles for Each Instruction

Instruction	Number of Cycles		Instruction	Number of Cycles
LDI	1		RET	4
IN	1		NOP	1
OUT	1		DEC	1
CALL	4		BRNE	2/1
SBI	2		RJMP	2
CBI	2			

```
.ORG 0
LDI R16,HIGH(RAMEND)
OUT SPH,R16
LDI R16, LOW(RAMEND)
OUT SPL,R16

SBI DDRB, 7

LOOP: CBI PORTB, 7
CALL DELAY
SBI PORTB, 7
CALL DELAY
RJMP LOOP

.ORG 0x50
DELAY: LDI R19, 50
LOOP2:  LDI R20, 200
LOOP1:  DEC R20
NOP
NOP
NOP
BRNE LOOP1
DEC R19
NOP
NOP
BRNE LOOP2
RET
```

a. **(9 Points)** Determine the time delay (in **microseconds**) for the instructions in the DELAY subroutine (given in gray). Assume that the crystal frequency is 16 MHz.

LOOP1: (1DEC+1NOP+1NOP+1NOP+2BRNE)200 – 1BRNE = 6(200)-1 = 1199 MCs = X

LOOP2: (1LDI+X+1DEC+1NOP+1NOP+2BRNE)50-1BRNE = (1205)50-1 = 60250-1 = 60249 MCs = Y

DELAY: 1LDI +Y+4RET = 60254 MCs = Z

Time required for DELAY = 60254 MC (1cc/1MC)(1s/16(10^6)cc) = 3765.875 (10^-6) s = 3766 us

b. **(4 Points)** Determine the time required (in **microseconds**) to execute one iteration through the loop LOOP.

LOOP: (2CBI+4CALL+Z+2SBI+4CALL+Z+2RJMP) = 120,522 MCs

Time required for 1 iteration of LOOP: 120,522 MC(1cc/1MC)(1s/16(10^6)cc) = 7532.625 us (or 7533 us)

c. **(3 Point)** From LOOP, what are the number of machine cycles when (PORTB bit 7) PB7 = 0 and when PB7?

Number of machine cycles when PB7 = 0:___2CBI+4CALL+Z = 60,260 MCs_____

Number of machine cycles when PB7 = 1:___2SBI+4CALL+Z+2RJMP = 60,262 MCs_____

3. **(12 Points)** Given the following sequence of code. Determine the values of the SP and registers in **hex** specified **after** each instruction is executed. The initial value of SP for this program sequence is SP = 0x0000 (given in the table below). Hint: remember the top location of data memory for the ATmega32.

Answers in hex

	SP	R16	R17	R18
	0x0000			
LDI R16, 0b00001111		0F		
LDI R17, $7F			7F	
LDI R18, 52				34
PUSH R18	085F			
PUSH R16	085E			
PUSH R17	085D			
POP R16	085E	7F		
POP R18	085F			0F
POP R17	0000		34	

0X0000	34
…	
0X085D	
0X085E	7F
0X085F	0F

4. Answer the following questions for the program sequence below.

```
LDI R20, 0xFE
LDI R21, 0x84
ADD R21, R20
```

```
      H
   1111 1
    1000 0100   (84 = R21)
  +1111 1110   (FE = R20)
1   1000 0010 (82 = R21)
C
```

a. **(5 Points)** Determine the values (**in hex**) of the General Purpose Registers R20 and R21 AFTER executing all of the instructions. Show your work for full credit.

R20 = _____0xFE_____ R21 = ____0x82_____

b. **(7 Points)** Determine the values of the C, Z, V, N, S and H flags in the SREG I/O register AFTER the program sequence has been executed.

C = ___1___ Z = ___0__ V = __0___ N = ___1___ S = _1 xor 0 = 1___ H = __1____

5. Answer the following question for the ATmega32.

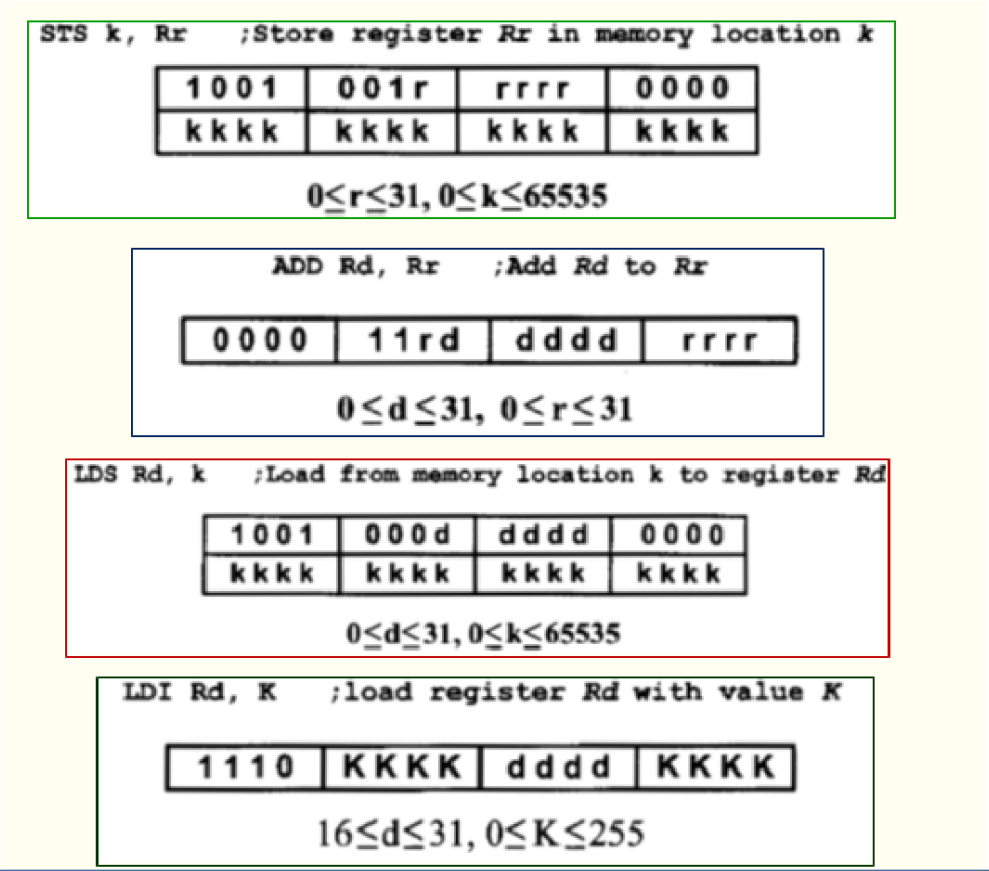
(**5 Points**) Write a sequence of assembly instructions and directives to initialize the constant TEMP to 0x67 using the .EQU directive, take the 1s comp of TEMP (flip the values of each bit of TEMP with the result NOT stored in TEMP), and store the result at data memory (SRAM) location 0x90.

```
.EQU TEMP = 0X67
LDI R16, TEMP
COM R16
STS 0X90, R16
```

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8. (14 Points) Given the following assembly language instruction sequence and machine codes for the instructions. In the table below, fill in the starting memory location and machine code for each instruction. Show your work for full credit. Fill in the Memory Locations and Machine Code with **HEX** values.

Starting Code Memory Location for Instruction (**HEX**)	Machine Code (see below for different instructions) (**HEX**)	Assembly Language Program Instruction
00000	E7F4	LDI R31, 0x74
00001	EA12	LDI R17, 0xA2
00002	0FF1	START: ADD R31, R17
00003	F7F1	BRNE START
00004	93F0 0352	STS 0x352, R31



BRNE BranchPlace; conditional branch -> branch if C = 0

1 1 1 1	0 1 K6 K5	K4 K3 K2 K1	K0 0 0 1

K6 K5 K4 K3 K2 K1 K0 = > -64 to +63 (2s complement range) offset with respect to the current PC

LDI R31, 0x74

31- 16 = 15 => 1111 = d

7 d 4
1110 0111 1111 0100 => E7F4

LDI R17, 0xA2

17-16 = 1 => 0001 = d

A d 2
1110 1010 0001 0010 => EA12

ADD R31, R17

R17 => 10001
R31 = 11111

0000 111 1 1111 0001 => 0FF1

BRNE START

PCnew = PCcurrent+1(1 code memory location)+k

00002 = 00003+1+k => k = -2 => 7 bit value (2s comp) => 0000010 (+2) => 1111101 + 1 = 1111110 (2s comp)

1111 01 1111110 001 => F7F1

STS 0x352, R31

R31 = 11111

1001 001 11111 0000 => 93F0
0X352 => 0000 0011 0101 0010 (16 BITS) => 0352